Thresholds That Travel: Adaptive Alarm Thresholds Let Attacks Through in ICS Anomaly Detection

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ABSTRACT

Anomaly detectors for industrial control systems (ICS) raise an alarm when a score crosses a threshold. The threshold is calibrated on clean data and must then hold a false-alarm budget for months, so deployments often let it adapt: they recalibrate it daily, track a running quantile, or refit an extreme-value tail as data arrive. We show that whatever an adaptive threshold learns from, an attacker can feed it, and that real attacks feed some of these rules without any adversarial effort.

We ran a pre-registered study on three datasets that no earlier analysis in our pipeline had opened (HAI 20.07, HyGrid-ICS and voraus-AD), with a world-model residual monitor and eleven threshold rules. On HAI 20.07's 38 attacks the frozen threshold detected 36 at 2.23 false-alarm episodes per day; a symmetric quantile tracker detected 20 (difference -0.42, 95% interval $[-0.58, -0.26]$) because earlier attacks had raised its threshold a median 12.9 times above calibration; fed only the attack samples, the tracker lost exactly as much. Post-hoc, the gap shrank to -0.16 ($[-0.37, 0.05]$) when an attack counted only if an alarm began inside it, because the frozen threshold was in alarm for 39% of HAI 20.07's unlabelled time. Daily recalibration and a ten times slower tracker also lost recall, and the same loss appeared on HyGrid-ICS and voraus-AD, on three earlier HAI releases, and with a second, model-free score. On attack-free development data, twelve two-minute nuisance alarms a day blinded every rule that learns from exceedances to a later, just-detectable setpoint attack; on the registered HAI 20.07 stream the real attacks had already done the same before any pulse arrived.

Streaming extreme-value thresholds (SPOT and DSPOT) refuse to learn from alarms and escaped this natural poisoning on the HAI data, but a closed-loop attacker who holds the score just under the current threshold drove their thresholds upward without raising one alarm and blinded them within an hour. A second registered study on the BATADAL water network replicated the tracker's loss (6 of 14 attacks against 14 for the frozen threshold, which was in alarm for 1.4% of the unlabelled time) and showed that slow, days-long attacks also inflate SPOT and DSPOT. Its registered primary rule nevertheless failed: the third hypothesis, that a closed-loop attacker blinds DSPOT, could not be tested because real attacks had already blinded DSPOT in all ten controls. A threshold that only tightens itself cannot miss an event the frozen threshold catches, on any input; its cost is false alarms when the plant drifts upward (2.43 times the budget on BATADAL, against 0.70 for the frozen threshold), while every rule that can raise its own threshold lost recall on at least one of seven datasets. A short analysis of the update rules explains the losses: a tracker calibrated to HAI 20.07's false-alarm rate needs 23 attack-free hours to forget one minute above its threshold. We argue that an alarm threshold may tighten itself but should loosen only with operator approval, and we release the registrations, ledger and code.

KEYWORDS

industrial control systems, anomaly detection, adaptive thresholds, poisoning, extreme value theory, pre-registration

1 INTRODUCTION

A process-monitoring anomaly detector has two parts: a score that grows when the plant behaves unlike its normal operation, and a threshold that turns the score into alarms. Most of the research effort goes into the score. The threshold is usually set once, on held-out normal data, to meet a false-alarm budget, and then expected to hold that budget on a plant that drifts: sensors age, seasons change load, and operators retune loops. When it does not hold, the standard remedies adapt the threshold online. Daily recalibration re-estimates a quantile on the last day of data. Online quantile tracking, the mechanism behind adaptive conformal inference [5], nudges the threshold up after every exceedance and down otherwise. Streaming peaks-over-threshold methods (SPOT and DSPOT [15]) refit a generalised Pareto tail as new values arrive, and the nonparametric dynamic threshold of Telemanom [9] recomputes its level on a trailing window.

Each of these rules learns from part of its input. Trackers and recalibration learn from the values that exceed the threshold. SPOT and DSPOT discard values above their alarm level but learn from values between a base level and the alarm level. A frozen threshold learns from nothing after calibration. We call the part of the score range a rule learns from its *learning region*, and the question of this paper is what happens when an attack, or an attacker, lands in it.

The answer depends on the region, and none of the adaptive rules we tested is safe:

- **Rules that learn from exceedances are poisoned by real attacks.** An attack is a long exceedance, so a tracker raises its threshold during the attack and stays high afterwards. The next attack then arrives under an inflated threshold. On fresh HAI 20.07 data the tracker missed 18 of 38 attacks, 16 of them attacks the frozen threshold caught. No adversarial planning is needed; the attacks in the recording did it.
- **The same rules can be poisoned on purpose and loudly.** Twelve two-minute pulses a day, the kind of nuisance alarm operators learn to acknowledge, raised these thresholds enough to hide a later attack that a frozen threshold caught every time (attack-free development data).
- **Rules that refuse to learn from alarms can be poisoned silently.** An attacker who reads the current SPOT or DSPOT threshold and holds the score just under it feeds every sample into the fitted tail. The tail extends, the threshold rises, the attacker follows it up, and no alarm is ever raised.

A threshold that only tightens automatically, and loosens only when an operator approves, cannot be pushed above the frozen threshold by any of the three, so it keeps the frozen threshold's recall by construction (Section 3.1). Its cost is extra false alarms when the plant drifts upward.

Method. The main study was registered before any test file, label or attack schedule of its three datasets was read. A ledger records every access to recorded attack data, the analysis harness refuses to run if any frozen source file changed after registration, and it computes the registered gates itself. The rules and the attack designs came from development on three earlier HAI releases whose attacks had already been used; we report how the claim changed during development, including two hypotheses we dropped. Extensions run after the registered finals are labelled exploratory.

Contributions.

(1) A registered comparison of eleven threshold rules on three never-opened datasets from three systems (a hardware-in-the-loop ICS testbed, a simulated hybrid power grid, and an industrial robot), with real-event recall, false-alarm rates and threshold trajectories (Section 6).
(2) Evidence that exceedance-learning thresholds lose recall on real attacks because earlier attacks inflate them, on seven datasets and two score families, with the mechanism visible in which attacks are missed, and a registered replication on the BATADAL water network (Sections 6 and 7).
(3) A learning-region analysis of adaptive thresholds: how long the tracker and daily recalibration remember an exceedance, as a function of the calibrated false-alarm rate alone (Section 3.1), and a matching attack for each class: supra-threshold pulse campaigns for exceedance learners and a closed-loop frog-boiling attack for extreme-value learners, including how the latter depends on the tail estimator (Section 7).
(4) A design rule, supported by these results, for deployed detectors: thresholds may tighten themselves; loosening needs a human (Section 8).
(5) The registration, access ledger, harness and verifier, in the anonymized artifact at https://anonymous.4open.science/.. ..

2 BACKGROUND AND RELATED WORK

ICS anomaly detection. Process-level detectors learn normal behaviour from sensor and actuator traces and flag deviations: residuals of a predictive model, reconstruction errors of an autoencoder, or statistical bounds. Public testbed recordings with labelled attacks (SWaT [6], WADI [1], BATADAL [16], and the HAI releases [14] for HAI 1.0 (20.07); later releases are documented in the dataset repository) are the usual benchmarks. Evaluations typically fix one threshold per detector, chosen on clean data or, less defensibly, on the test labels. Wu and Keogh [17] list further benchmark pitfalls.

Threshold rules. Static thresholds come from a quantile or a false-alarm budget on clean data. Adaptive variants recalibrate on a trailing window, track a target exceedance rate online [5], fit an extreme-value tail and update it per observation (SPOT, and DSPOT for drifting streams [15]), or recompute a level on a trailing window of smoothed errors [9]. Alarm-management practice in process plants treats nuisance and chattering alarms as a condition to eliminate, with explicit alarm-rate targets [10]; operators who acknowledge such alarms without investigating them are the failure mode our pulse campaigns exploit.

Poisoning of anomaly detectors. Rubinstein et al. [13] showed that slowly shifted traffic ("boiling frog" poisoning) can stop a PCA-based network anomaly detector from flagging a later attack. Kloft and Laskov [11] analysed poisoning of online centroid detectors. Chan-Tin et al. [3] described the frog-boiling attack on outlier rejection in network coordinate systems, where every update stays within the rejection threshold while the system drifts. In ICS, Kravchik and Shabtai [12] found that autoencoder detectors retrained online on SWaT resisted their poisoning attacks, and Erba et al. [4] concealed attacks from reconstruction-based detectors by manipulating sensor readings. Hong [7] studied gradual drift against world-model monitors in simulation, without attacks or adaptive thresholds. These works target the model or the score. We target the threshold rule and show that it can fail even when the model resists poisoning.

Distribution shift and calibration. Adaptive conformal inference keeps a long-run miscoverage rate under arbitrary shift, a guarantee about the rate of exceedances. That guarantee holds whether the exceedances come from benign drift or from attacks, which is the property we exploit in Section 7.

3 THREAT MODEL AND LEARNING REGIONS

System. A monitor computes a score s_t from plant telemetry and raises an alarm when $s_t > \tau_t$. The rule that sets τ_t is calibrated once on clean data to a false-alarm budget and then runs online. Operators acknowledge alarms; persistent nuisance alarms are acknowledged without investigation.

Learning region. For a rule R, the learning region $L(R, \tau_t)$ is the set of score values that change R's future thresholds. Table 1 gives the rules we study.

A rule can loosen (raise τ_t) only through values in the part of its learning region that pushes upward. That gives three attacker classes, from weakest to strongest:

- **A0, no attacker.** Real attacks are themselves exceedances. A rule whose upward learning region includes exceedances is poisoned by the attacks it is meant to detect.
- **A1, open-loop injection.** The attacker can perturb one control loop (setpoints or sensor readings) without seeing the monitor. Supra-threshold pulses land in the exceedance region; sub-alarm pulses land in the band.
- **A2, closed-loop.** The attacker can read τ_t (white-box, or by probing) and shape the score. Holding s_t at $0.97\,\tau_t$ places every sample at the top of the band with no alarm. We evaluate A2 at the score level, an upper bound on what a physical attacker can do, and through noise spoofed onto three sensor readings of one loop, scored by the real monitor.

Table 1: Update and learning region of each threshold rule.

Rule	Update	Learning region
frozen, static margin	none	empty
symmetric tracker $q_{t+1} = q_t + \eta(\mathbf{1}[s_t > q_t] - p)$	up on exceedance, down otherwise	all values; upward moves only from $s_t > \tau_t$
daily recalibration	$(1-p)$ quantile of the last day	the top p of each day, whatever its source
SPOT	adds $s_t - t$ to the tail fit when $t < s_t \le z_q$	the band $(t, \tau_t]$
DSPOT	as SPOT on s_t minus a moving mean of non-alarm values	the band, plus every non-alarm value through the mean
tighten-only	min(ceiling, tracker); the ceiling rises only through a permit	downward only, unless a permit fires

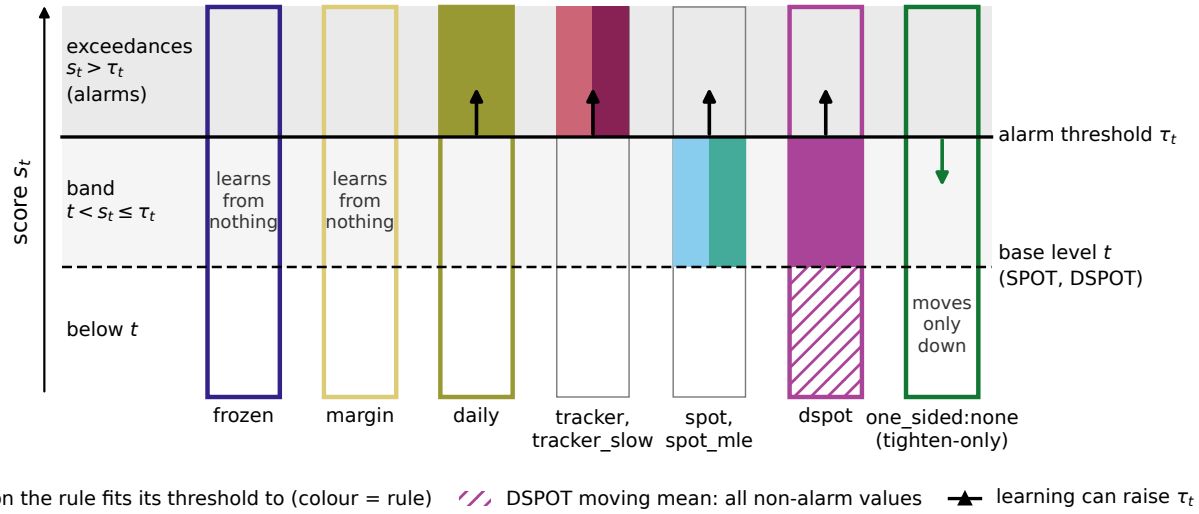


Figure 1: Learning regions of the threshold rules (schematic). The alarm threshold τ_t and, for SPOT and DSPOT, the base level t below it split the score range into exceedances ($s_t > \tau_t$), the band ($t < s_t \le \tau_t$) and values below t. A filled column segment marks the region a rule fits its threshold to; DSPOT's moving mean also absorbs every non-alarm value (hatched). Daily recalibration and the trackers learn from exceedances, so the attacks they are meant to detect raise their thresholds; frozen and the static margin learn from nothing; one_sided:none can only move downward. Data status: schematic with no data, so the registered / exploratory / post-hoc label does not apply; the rule behaviour follows Table 1.

The attacker's goal is a terminal attack that the frozen threshold would detect but the adapted threshold does not.

3.1 How long a threshold remembers

Three facts follow from the update rules alone. They predict most of what Sections 6 and 7 measure.

Proposition 1 (tracker memory). Under $q_{t+1} = q_t + \eta(\mathbf{1}[s_t > q_t] - p)$, each step with $s_t > q_t$ raises the threshold by $\eta(1-p)$ and each other step lowers it by ηp. A rise built from E exceedance steps therefore takes $E(1-p)/p$ quiet steps to undo, whatever the step size η.

The calibrated exceedance rate p sets the memory. On HAI 20.07 ($p = 7.2 \times 10^{-4}$ per second) the ratio is 1,385, so one minute above the threshold keeps it raised for 23 hours. On BATADAL ($p = 0.0075$ per 15-minute step) it is 133, so one step above the threshold keeps it raised for 33 hours. A slower tracker (tracker_slow, $\eta/10$) climbs less during a short attack but needs the same 1,385 quiet seconds per exceedance second to come back down, which is why it lost recall too (Section 6.2). Every recorded change of the tracker on the HAI 20.07 and BATADAL test streams is a whole number of these up and down steps (largest deviation 7×10^{-6} of a step), and from the levels the tracker reached it would have needed 43 to 209 days of attack-free data to return to τ_0.

Proposition 2 (daily memory). Daily recalibration sets the next day's threshold to the $(1-p)$ quantile of the last day's N steps. An attack that holds the score above a level a for more than $pN+1$ steps of one day lifts the next day's threshold to at least a.

That is 64 seconds on HAI 20.07 and two 15-minute steps on BATADAL. The memory is one day, so daily recalibration is poisoned whenever attacks fall on consecutive days, as they do in the HAI test files, and it still catches an attack whose first day follows a quiet one: on BATADAL, whose attacks last one to five days and begin 1.7 to 24 days after the previous one ends, it caught 14 of 14 with its threshold at 0.69 τ_0 at the median onset.

Proposition 3 (tighten-only). If $\tau_t \le \tau_0$ at every step, every sample that alarms under the frozen threshold also alarms under τ_t. The rule therefore detects every event the frozen threshold detects, on any input and against any attacker, adaptive ones included. What it can lose is false alarms.

Proposition 3 means the tighten-only results below measure its cost, not its recall. On HAI 20.07, HyGrid-ICS and voraus-AD its recall and false alarms equalled the frozen threshold's: the inflated tracker under the cap kept it at τ_0 (on HAI 20.07 it sat below τ_0 in 0.4% of minutes). Its own behaviour shows on BATADAL, where the tracker sat below τ_0 for 3.2% of the test time and the rule ran at 2.43 times the false-alarm budget against 0.70 for frozen, and on a development split of HAI 23.05 (5.98 against 4.05).

4 MONITOR, RULES AND DATA

Monitor. D1 is the world-model residual monitor of earlier registered studies on HAI [anonymized]: an action-conditioned ridge regression with LightGBM residual corrections (60 trees, history 3) predicts each sensor one and five steps ahead; the score is the mean of the three largest standardised absolute residuals at each horizon, summed and smoothed over 60 s. D0 is a model-free control: the mean of the three largest standardised first differences. Both are calibrated on the most recent 20% of normal training data: to 2 false-alarm episodes per day on the 1 Hz stream datasets (episodes merged across 60 s gaps), to a per-sample false-alarm target of 0.05 on voraus-AD, and to one false-alarm episode per week on BATADAL (episodes merged across 1 h).

Rules. frozen (τ_0 at the budget); margin (τ_0 + 0.25× the spread between the calibration 99.9th percentile and median); daily (trailing-day quantile at the calibration exceedance rate p); tracker (η = 0.01× spread on the stream datasets, 0.05× spread on voraus-AD) and tracker_slow ($\eta/10$, the best false-alarm adherence among trackers in development); tighten-only with four automatic permits and an oracle permit that simulates an operator who refuses to loosen while an event is under way: eleven rules in all. On voraus-AD the spread is the calibration 99th percentile minus the median and daily recalibration happens every 50 samples. The streaming extreme-value rules (SPOT and DSPOT, with method-of-moments and maximum-likelihood tail fits [8, 15]) were not part of the tt1 registration: they enter in Section 7 as exploratory and post-hoc analyses and are registered only in the BATADAL replication (tt1b).

Registration. The protocol fixed the monitor, splits, rules, injection designs, gates and seeds; the harness recorded one ledger entry per dataset before reading a test array, refused to run if a frozen file's SHA-256 differed from the registration, and computed the gates from its own output. A verifier checks that each access followed the registration commit, that each dataset was opened once, and that the stored gates match a recomputation (27 of 27 checks pass).

5 HOW THE HYPOTHESIS CHANGED DURING DEVELOPMENT

All development used normal training data and the already-opened attack files of HAI 21.03, 22.04 and 23.05. We report it because the registered claim differs from the one we set out to test.

(1) *Tighten-only with automatic loosening.* Our starting design let a threshold tighten itself and loosen only through a statistical permit. A slow sub-threshold drift ("boiling frog" in the style of Rubinstein et al.) did not poison any rule: the world model absorbed the drift. Supra-threshold pulses did inflate the trackers, but every automatic permit we tried either opened under the pulses (spread permit: 3 of 30 terminal attacks caught in development round r3, 5 of 30 in round r4) or failed to follow benign drift (shift permit).
(2) *Static safety margin.* A margin at half the calibration exceedance rate cut the false-alarm rate on pseudo-deployment splits without always reaching the budget (HAI 20.07 development: 3.79 episodes per day against 4.42 for frozen, budget 2), and cost 4 to 10 points of recall on the spent attacks (interval lower bounds down to −0.17), so we registered a smaller margin and made it secondary.
(3) *Frozen versus adaptive.* On the spent attacks the frozen threshold held the budget on all three real test periods (0.66, 1.48 and 1.27 episodes per day), while the tracker lost 22, 31 and 37 points of recall with intervals entirely below zero. This became the registered primary.

6 REGISTERED RESULTS

6.1 Primary: HAI 20.07

False-alarm episodes per day are counted over the 4.94 days of unlabelled time inside the 5.15-day test streams. G1 passes as registered, but on this stream the pulses were not needed: the adaptive rules also missed the terminal step in the control runs, because the test stream's real attacks had already raised their thresholds 7 to 26 times above τ_0. The pulse effect in isolation comes from development on attack-free normal data (frozen 30/30; daily 4/30, control 21/30; tracker 1/30, control 23/30; tracker_slow 3/30, control 25/30).

6.2 Real attacks inflate adaptive thresholds

At HAI 20.07 attack onsets the median threshold stood at 25.6 τ_0 for daily recalibration, 12.9 for the tracker and 6.4 for the slow tracker. HyGrid's daily rule equals frozen because its 14.5 h final never completes a day.

Time in alarm and onset recall (post-hoc). Episode counts hide how long an alarm lasts. On HAI 20.07 the frozen threshold was

Table 2: Datasets. Event counts are the labelled events of each final split.

Dataset	System	Training data	Final data	Events
HAI 20.07 (primary)	HIL testbed (boiler, turbine, water treatment)	547,624 s normal, 31 sensors, 20 actions	test1 + test2, 5.15 days	38 attacks
HyGrid-ICS	simulated hybrid grid SCADA, 5 Hz to 1 Hz	9.6 h (label-blind first 40%)	14.5 h	91 attacks
voraus-AD	industrial robot, 100 Hz to 10 Hz	948 normal samples	1174 samples	755 anomalous samples
HAI 21.03, 22.04, 23.05 (development)	HIL testbed	normal training data	spent test files	50, 58, 52 attacks
BATADAL (registered replication)	water distribution network, 15 min	384 days normal	two test files, 260 days	14 attacks

Table 3: Registered gates of tt1 on HAI 20.07. Data status: registered.

Gate	Registered test	Result
G1	after a pulse campaign (12×120 s per day at twice the terminal magnitude), daily, tracker and tracker_slow detect the terminal step in at most (frozen − 0.5) of placements; frozen at least 0.8	frozen 10/10, each adaptive rule 0/10: pass
G2	recall(tracker) − recall(frozen) on the real attacks, 95% paired event bootstrap upper bound below 0	0.526 vs 0.947, −0.42 [−0.58, −0.26]: pass
G3	frozen false-alarm episodes per day on unlabelled test time at most 4	2.23: pass (above the 2.0 budget)

Table 4: Event recall and the paired difference against the frozen threshold (95% paired event bootstrap) on the D1 score. Data status: registered (tt1).

Dataset (events)	frozen	daily	tracker	tracker_slow	tighten-only
HAI 20.07 (38 attacks)	0.947	0.474 [−0.63, −0.32]	0.526 [−0.58, −0.26]	0.711 [−0.37, −0.11]	0.947
HyGrid-ICS (91 attacks)	0.297	0.297	0.154 [−0.22, −0.08]	0.165 [−0.21, −0.07]	0.297
voraus-AD (755 anomalous samples)	0.334	0.148 [−0.21, −0.16]	0.082 [−0.28, −0.22]	0.213 [−0.14, −0.10]	0.334

in alarm for 39% of the unlabelled time: the D1 score sat above τ_0 for the first 23.8 h of test1 and for most of a 17-hour stretch of test2, and each such run counts as one false-alarm episode. Ten of the frozen threshold's 36 detections were attacks that began while one of these alarms was already running. Counting an attack only when an alarm episode starts inside it, the frozen threshold detects 26 of 38, the tracker 20 (−0.16, [−0.37, 0.05]), the slow tracker 27 (+0.03, [−0.18, 0.24]) and daily recalibration 11 (−0.40, [−0.55, −0.24]). On BATADAL the frozen threshold was in alarm for 1.4% of the unlabelled time and the stricter count changes nothing (tracker −0.57, [−0.79, −0.29]). On HyGrid-ICS (4.1% in alarm) the 91 attacks cover 83% of the final rows, about 100 s apart on average, so one merged alarm episode spans several attacks and the stricter count credits few of them: frozen 3 of 91, tracker 8 (+0.05, [0.00, 0.11]), slow tracker 5. There the stricter count reverses the sign, but it measures how alarms merge across back-to-back attacks more than what the threshold detects. G2 stands as registered, but the tracker's HAI 20.07 gap depends on crediting the frozen threshold's long alarms; the BATADAL gap does not.

Table 5: False alarms. Data status: registered (tt1).

Dataset	frozen	margin	daily	tracker	tracker_slow
HAI 20.07 (episodes/day)	2.23	2.23	0.20	1.01	2.23
HyGrid-ICS (episodes/day, 2.5 h unlabelled)	19.2	0.0	19.2	19.2	19.2
voraus-AD (per-sample FPR, target 0.05)	0.081	0.033	0.086	0.012	0.014

6.3 False alarms

Where an adaptive rule raised fewer false alarms than frozen, it bought them with the recall losses of Section 6.2. The one rule that cut false alarms at no recall cost here, the static margin on HyGrid-ICS (0.0 against 19.2 per day at identical recall), does not adapt. The margin matched frozen recall on HAI 20.07 and held voraus-AD's target at a cost of 12 points of recall there. Counted in time rather than episodes (post-hoc), the frozen threshold was in alarm for 39%

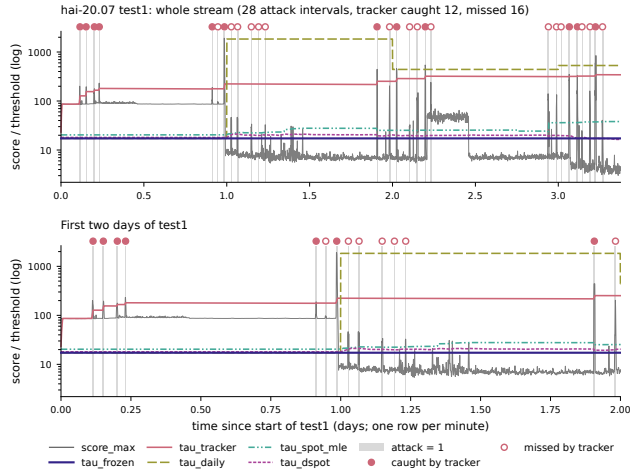


Figure 2: Threshold trajectories on the HAI 20.07 test1 stream (4,860 minutes). Grey: the D1 score score_max. Coloured: the frozen, tracker, daily, spot_mle and dspot thresholds, log scale. Grey shading marks attack-labelled minutes; each marker above the panel is one attack interval, filled if the tracker caught it. test1 holds 28 of the 38 attacks; the tracker caught 12. The score stays above $\tau_0 = 17.26$ for the first 23.8 h of test1 with no attack label, so the frozen threshold is in alarm throughout that stretch; this unlabelled stretch lifted the tracker to 5.25 τ_0 before the first attack, and the early attacks then raised the tracker and daily thresholds far above frozen. Later attacks arrive under the inflated thresholds. Data status: post-hoc analysis; the trajectories were computed after the registered finals. The recomputed calibration threshold equals the registered one exactly. Over test1 and test2 the traces give frozen 36/38 and tracker 20/38 (registered tt1) and spot_mle 35/38 and dspot 36/38 (post-hoc: SPOT and DSPOT were not in the tt1 registration). test2 is not drawn; test1 has no missing values.

of the unlabelled time on HAI 20.07, 4.1% on HyGrid-ICS and 1.4% on BATADAL; the tracker for 0.2%, 0.3% and 0.4%.

7 LEARNING REGIONS UNDER ATTACK (EXPLORATORY)

The analyses in this section were not registered. They use the spent HAI releases for development and reopen the three tt1 datasets after their finals; every access is in the ledger.

7.1 Extreme-value thresholds escape natural poisoning on the HAI data

The exceptions: on HyGrid-ICS the moments SPOT lost 9 points ($[-0.15, -0.03]$); on voraus-AD, 190 calibration samples leave about four tail points above SPOT's base level (the calibration 98th percentile), and SPOT ran at 32% false positives against a 5% target.

7.2 The effect does not depend on the score

With D0 the tracker lost recall on every dataset where it was measured: $[-0.38, -0.14]$, $[-0.21, -0.05]$ and $[-0.19, -0.02]$ on the spent releases, $[-0.32, -0.08]$ on HAI 20.07, $[-0.12, -0.02]$ on HyGrid-ICS, $[-0.04, -0.01]$ on voraus-AD and $[-0.86, -0.36]$ on BATADAL.

7.3 Mechanism: attacks that follow attacks

The tracker's misses concentrate on attacks that start within 6 h of the previous attack's end. D1 recall on close versus isolated attacks was 0.75 / 1.00, 0.41 / 1.00 and 0.20 / 0.67 on the spent releases and 0.48 / 0.71 on HAI 20.07 (31 close, 7 isolated). The frozen threshold also detected close attacks less often on two releases, so part of the gap reflects harder attacks; the tracker's gap is larger on every release.

Attacks are what raise the tracker. Rerun post-hoc with its update fed only the attack-labelled samples (the others replaced by the calibration median for the update, alarms still on the real score), the tracker detected 20 of 38 on HAI 20.07 and 14 of 91 on HyGrid-ICS, exactly its registered counts, and 7 of 14 on BATADAL (6 registered). Fed only the unlabelled samples, it detected 28 of 38, 27 of 91 (equal to frozen) and 11 of 14: the unlabelled exceedances of HAI 20.07 and BATADAL add a smaller loss of their own.

7.4 Loud poisoning: pulse campaigns

Normal data of the spent releases, 30 placements; the terminal step is the smallest setpoint step the frozen threshold detects on the clean stream.

Open-loop sub-alarm pulses did not poison SPOT: its threshold fell (0.82 to 0.98 τ_0), and many pulses crossed SPOT's own alarm level and were refused.

7.5 Silent poisoning: the closed-loop frog

Score-level attacker A2, normal data of the spent releases, 10 placements each: the attacker holds the score at 0.97 τ_t, then, after a one-hour lead, a 600 s bump at 1.05 τ_0 follows (600 s is the terminal length of every development campaign).

Each rule is compared with its own control (the same bump without a campaign). The frozen and both SPOT controls caught 30/30 and DSPOT's 24/30; the tracker and daily controls already missed some bumps on the drifting releases, and the campaign did not lower them further. After 1 h the moments SPOT threshold stood at 1.1 to 4.5 τ_0; after 24 h it had grown by more than five orders of magnitude, and the MLE fit left the floating-point range on one release. A synthetic control with a light tail (the 60-step moving average of $|N(0,1)|$, five seeds) did not run away: the moments SPOT threshold stayed at 0.95 to 0.97 τ_0 and caught every terminal bump, and maximum-likelihood SPOT rose only to 1.02 to 1.04 τ_0. The runaway therefore depends on the heavy tail of real residual scores.

A physically realizable version. The score-level attacker sets the score directly. We also gave the attacker only the readings of three sensors coupled to one setpoint channel. Every 5 minutes it reads the rule's threshold and spoofs the largest zero-mean noise whose D1 score, computed by the real world model on the spoofed history,

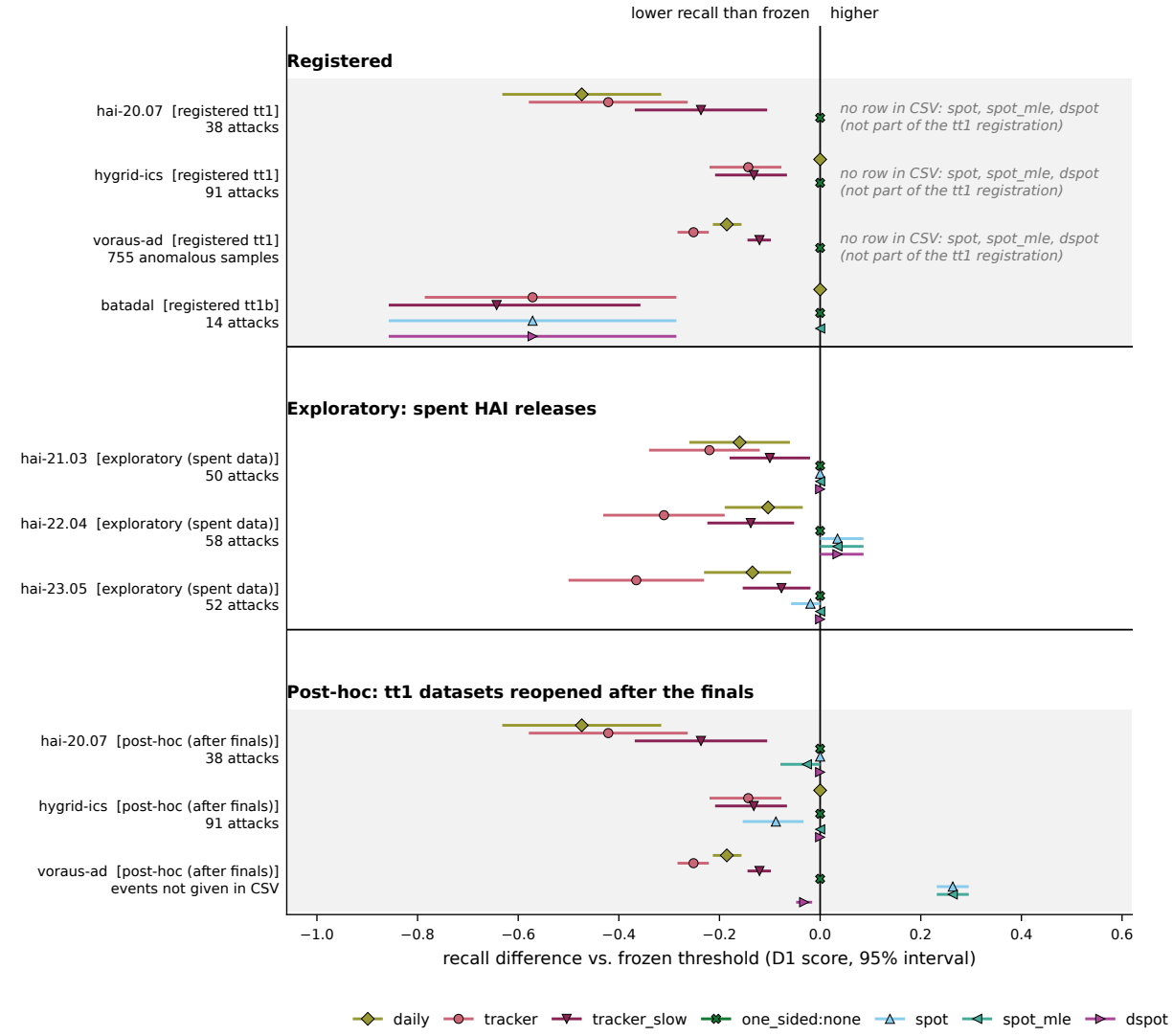


Figure 3: Paired difference in event recall between each adaptive rule and the frozen threshold on D1, with 95% intervals. Values left of 0 mean lower recall than frozen. Three blocks, each row labelled with its data status: (1) registered results from tt1 (hai-20.07, hygrid-ics, voraus-ad) and tt1b (batadal); (2) exploratory development results on the spent HAI releases; (3) post-hoc results on the tt1 datasets, reopened after the finals. spot, spot_mle and dspot have no rows for the registered tt1 datasets because those rules were not part of the tt1 registration; the gaps are labelled and the post-hoc block supplies the comparisons. one_sided:none equals frozen (0 $[0, 0]$) in every block. The positive spot and spot_mle values for voraus-ad (post-hoc) are not a gain: with 190 calibration samples SPOT ran at a per-sample false-positive rate of 0.32 against the 0.05 target (Section 7.1).

stays at 0.97 of that threshold; after 6 h it stops, and 1 h later the just-detectable setpoint step follows. This attacker is white-box: it can run the monitor.

Controls caught 12 of 12 for frozen and both SPOT variants and 9 of 12 for DSPOT. On HAI 22.04 both SPOT variants missed the terminal step on all four placements, and the campaign raised no alarm. The spoofed noise had a scale of 0.02 to 0.04 robust units against frozen and the SPOT variants, and 0.05 to 0.12 against DSPOT, whose moving mean follows the noise-driven rise, so the attacker's cap rises with the mean rather than with the tail.

7.6 Registered replication on BATADAL

We registered a second study (tt1b) to test the exploratory findings of Sections 7.1 and 7.5 on data no analysis here had opened: the C-Town network of BATADAL [16], whose original attacks conceal themselves in the SCADA readings, in the re-simulation without that concealment published with the artifact of Erba et al. [4], put

Table 6: Event recall of the extreme-value rules. Data status: exploratory on the spent releases, post-hoc on HAI 20.07.

Data	frozen	tracker	SPOT (moments)	SPOT (MLE)	DSPOT
HAI 21.03 / 22.04 / 23.05	1.00 / 0.76 / 0.60	0.78 / 0.45 / 0.23	1.00 / 0.79 / 0.58	1.00 / 0.79 / 0.60	1.00 / 0.79 / 0.60
HAI 20.07 (post-hoc)	0.947	0.526	0.947	0.921 [−0.08, 0.00]	0.947

Table 7: Terminal attacks caught after a pulse campaign (control in parentheses). Data status: exploratory (normal data of the spent releases).

Campaign	frozen	daily	tracker	tracker_slow	SPOT	DSPOT
12 × 120 s supra-threshold pulses per day	30/30 (30/30)	4/30 (21/30)	1/30 (23/30)	3/30 (25/30)	29/30 (30/30)	23/30 (25/30)
24 × 300 s sub-alarm pulses per day, open loop	30/30	23/30 (23/30)	19/30 (21/30)	23/30	29/30	22/30 (24/30)

Table 8: Terminal attacks caught after a closed-loop frog campaign, 30 placements. Data status: exploratory (normal data of the spent releases).

Rule	24 h campaign	6 h	1 h	open loop, fixed level, 24 h
frozen, tracker_slow, tighten-only	as control	as control	as control	as control
tracker	as control	as control	as control	16/30 (control 20/30)
SPOT (moments)	0/30	0/30	0/30	not moved; alarms throughout
DSPOT	0/30	0/30	0/30	not moved; alarms throughout
SPOT (MLE, refit every 2048 peaks)	8/30	8/30	13/30	11/30

Table 9: The physically realizable frog. Data status: exploratory (normal data of the spent releases).

Rule	Terminal caught after campaign / in control (12 placements)	Threshold at the terminal step (median per release)
frozen	12 / 12	1.0 τ_0
SPOT (MLE)	4 / 12	1.04 to 1.59 τ_0
SPOT (moments)	6 / 12	0.78 to 1.42 τ_0
DSPOT	9 / 9	0.88 to 1.12 τ_0

on a 15-minute grid. All time-scale settings were fixed in physical time before access: 1 h smoothing, episodes merged across 1 h, one false-alarm episode per week, a one-day DSPOT mean. The test files hold 14 attacks in 260 days.

We retain the primary rule as registered and report H3 as uninformative rather than refuted. SWaT and WADI are registered under the same protocol and were requested on 2026-10-23; they run once their loaders are added by amendment, and the replication claim is the set of per-dataset outcomes.

H3 failed because real attacks had already blinded DSPOT: its threshold stood 24 times above τ_0 at attack onsets, and it detected 6 of 14 attacks. BATADAL attacks last days, and the smoothed score spends long stretches in the band below the alarm level, which is exactly where moments SPOT and DSPOT learn.

With the model-free D0 score, MLE SPOT also lost recall (9/14, $[-0.64, -0.14]$). The margin and the tracker step are scaled by the calibration spread (99.9th percentile minus median); BATADAL's residual scores are heavy-tailed, so the margin sat 173 times above τ_0. Parameters scaled by a tail spread are fragile when the tail is heavy.

8 DESIGN GUIDANCE

A threshold rule's learning region is its attack surface. Every automatic loosening path we tested could be pushed upward: exceedance tracking and daily recalibration by real attacks and by loud pulses; tail refitting and moving-mean drift tracking by a closed-loop attacker; the spread and shift permits by pulses and by benign drift. The rules that held were the ones that never loosen themselves: frozen, and tighten-only without a permit or with an operator-gated one. A static margin never loosens either, but it is not safe by construction: on BATADAL its spread-scaled offset sat 173 times above τ_0 and it caught 3 of 14 attacks, the largest recall loss we measured.

We therefore recommend:

(1) **Let thresholds tighten automatically, never loosen.** A tracker capped at the calibrated threshold cannot lose an event the frozen threshold catches (Proposition 3) and can only add sensitivity in quiet periods.
(2) **Route loosening through an operator** who confirms the plant is in normal operation, and treat a request to loosen during or soon after an alarm as suspect.
(3) **Recalibrate offline on verified clean data**, as a change-controlled event, when drift raises false alarms; our frozen thresholds held 0.66 to 2.23 episodes per day on four HAI test periods, but on HAI 20.07 the threshold was in alarm

Table 10: Registered hypotheses of tt1b on BATADAL. Data status: registered.

Hypothesis	Registered test	Result
H1	recall(tracker) − recall(frozen), interval upper bound below 0	6/14 vs 14/14, −0.57 [−0.79, −0.29]: pass
H2	recall(MLE SPOT) − recall(frozen), lower bound at least $-\max(0.10, 2/N)$	14/14, difference 0: pass
H3	24 h closed-loop frog: DSPOT at least 0.5 below its control; frozen and tighten-only within 0.1	DSPOT already 0/10 in the control: fail
Primary rule	H1 and H2 and H3	fail

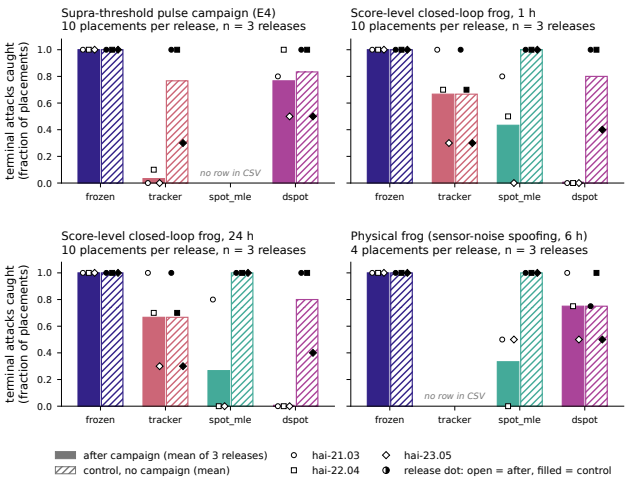


Figure 4: Terminal attacks caught after each campaign (solid) and in the matching control (hatched), for frozen, tracker, spot_mle and dspot. Each bar is the mean over the three spent HAI releases; dots are the per-release values, and every release had the same number of placements, so the mean equals the pooled fraction. Panels: the supra-threshold pulse campaign (E4, 10 placements per release); the score-level closed-loop frog at 1 h and 24 h (10 placements); and the physical frog, which spoofs noise on three sensors for 6 h (4 placements). Data status: all four panels are exploratory results on the normal data of the spent releases. The source CSV has no spot_mle row for E4 and no tracker row for the physical frog; both slots are left empty and labelled.

for 39% of the unlabelled time, the kind of drift an offline recalibration should catch.

(4) **Watch the threshold.** Log τ_t; a threshold that rises during an incident, or climbs steadily without alarms, is itself an indicator.

(5) **Budget alarm time as well as episodes.** An episode budget let a 24-hour alarm on HAI 20.07 count once, and an attack that begins inside a running alarm gives the operator nothing new to see.

The cost is false alarms under benign upward drift: on a development split of HAI 23.05 the tighten-only rule ran at 5.98 times the budget against 4.05 for the frozen threshold. A static margin can cut that cost on some plants (HAI 23.05 development: 0.58 times the budget) but costs recall on others (voraus-AD: 12 points; BATADAL: 11 of 14 attacks), and should not be used where the calibration spread is set by a heavy tail.

Table 11: BATADAL, D1 score. Data status: registered (tt1b).

Rule (D1)	Recall	False alarms (× budget)	Onset inflation
frozen	14/14	0.70	1.0
tighten-only	14/14	2.43	1.0
SPOT (MLE)	14/14	0.81	1.0
daily	14/14	6.34	0.69
tracker	6/14	1.83	409
tracker_slow	5/14	0.70	74
SPOT (moments)	6/14	0.13	26
DSPOT	6/14	0.13	24
margin	3/14	0.05	173

9 LIMITATIONS

- Two registered studies with one primary dataset each; HAI 20.07 shares a testbed with the development releases, and BATADAL has 14 attacks.
- HyGrid-ICS is a single simulated day with 2.5 h of unlabelled final time; voraus-AD anomalies are not attacks. Post-hoc, 0.15% of HyGrid-ICS's label-blind development rows (251 short runs) carry an attack label, against 83% of its final rows, so its calibration is very slightly contaminated.
- The pulse campaigns and both frogs are synthetic. The score-level frog sets the score directly and reads τ_t; the physically realizable version also runs the monitor itself on the spoofed history to choose each noise scale. Both are white-box upper bounds.
- SPOT's behaviour depends on the tail estimator and on the score's tail shape; we report both.
- Parameters scaled by a calibration spread (the static margin and the tracker step) are fragile on heavy-tailed scores: on BATADAL the margin sat 173 times above τ_0 and missed 11 of 14 attacks.
- Event recall counts any alarm inside a labelled interval, so it credits an attack that falls inside an alarm already running; on HAI 20.07 that gave the frozen threshold 10 of

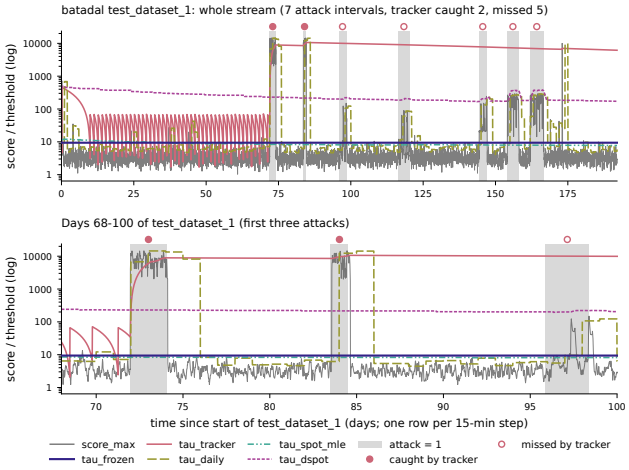


Figure 5: Threshold trajectories on the BATADAL test_dataset_1 stream (18,433 steps of 15 minutes, 192 days); layout as in Figure 2. test_dataset_1 holds 7 of the 14 attacks; the tracker caught 2. The lower panel zooms to days 68 to 100, which cover the first three attacks, because the first attack starts on day 72. score_max is missing for the first 10 steps ($t = 0$ to 9, 2.5 h); those steps are left as a gap and the thresholds are drawn there as given. The days-long attacks lift the tracker to about 10^4 and keep DSPOT's threshold far above τ_0 (9.39), while frozen and spot_mle stay near τ_0. Data status: post-hoc analysis of the tt1b dataset, which was opened once for the registered study; the recomputed τ_0 equals the registered value exactly. Over both test files the traces give frozen 14/14, tracker 6/14, spot_mle 14/14 and dspot 6/14, the registered tt1b counts. test_dataset_2 is not drawn.

its 36 detections (Section 6.2). We do not model operator response time.

- Our monitor is one world-model residual and one model-free score; reconstruction-based detectors were not tested.
- No claim of runtime authority, deployment safety or incident prevention follows from these results.

10 CONCLUSION

On four never-opened datasets, adaptive thresholds that learn from exceedances missed attacks that a frozen threshold caught, because earlier attacks had raised them; fed only the attack samples, the tracker lost as much as with everything. On HAI 20.07 part of the gap came from long alarms of the frozen threshold that also covered later attacks; on BATADAL none of it did. Extreme-value thresholds avoided that failure on the HAI data but not on days-long BATADAL attacks, and an attacker who stayed under them could drive them upward without an alarm. How long a tracker or a daily recalibration stays raised follows from its calibrated false-alarm rate alone, so making the tracker slower does not make it forget faster. A threshold that only tightens itself keeps the frozen threshold's recall by construction; on the seven datasets it caught the same events, and on BATADAL, the one registered dataset where its tightening changed the alarms, it ran at 2.43 times the false-alarm budget against 0.70 for the frozen threshold.

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ETHICAL CONSIDERATIONS

All experiments ran offline on public recordings: HAI 20.07 to 23.05, HyGrid-ICS, voraus-AD and the BATADAL re-simulation, each used under its publisher's terms. No live plant, testbed, vendor product or operator was involved, and no human subjects took part. The pulse campaigns and both frog attacks were injected into recorded normal data or into the score; the physically realizable frog changes recorded sensor values in software and is scored by our own monitor.

The attacks are not new in kind. Boiling-frog poisoning of self-updating detectors has been public since 2009 [3, 13], and the threshold rules we attack are textbook algorithms. What the paper adds for an attacker is the observation that an adaptive threshold widens the window for a later attack; what it adds for a defender is a rule that closes that window at a known false-alarm cost. We judged that publishing both helps defenders more than attackers, because the defence is a configuration change (cap the threshold at its calibrated value) that any operator of an adaptive threshold can apply today. The released code generates attacks only against recorded data and a score file; it contains no tooling for reaching a control network.

OPEN SCIENCE

The two pre-registrations, the data-access ledger, the attack and threshold harnesses, the registered-gate verifiers and the figure data and plotting code are available in the anonymized artifact at https://anonymous.4open.science/..., which also contains the result files behind every number in this paper and the verifier reports (27 of 27 checks for tt1, 8 of 8 for tt1b). The raw datasets are not redistributed here and are available from their publishers: HAI from its dataset repository, BATADAL from Taormina et al. [16] and the no-concealment re-simulation from the artifact of Erba et al. [4], voraus-AD from Brockmann et al. [2], and HyGrid-ICS from its Zenodo record [18]. Reproducing the registered results requires one access to each raw dataset under the access rules recorded in the ledger.